Cinematic Data Retrieval: Pursuing a Better Search Mechanism

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Figure 1: Movies And Series

Abstract

The rapid growth of digital media has increased the need for effective information organization and retrieval systems. This project addresses this challenge in the context of the movie and television domain, integrating structured and unstructured data from multiple sources. This report describes the construction of a reproducible data processing pipeline, including data cleaning, integration, and quality assessment, forming the basis for the development of a search engine for movies and series.

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CCS Concepts

• **Information systems → Information retrieval**.

Keywords

Movies, Series, Cinema, Film, Reviews, Dataset, Search Engine, Pipeline, Data Retrieval, Data Preparation, Data Analysis, Data Processing, Data Refinement

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1 Introduction

This project's primary goal is to apply the principles and techniques of information processing to a real-world domain, exploring the challenges of collecting, organizing, and analyzing heterogeneous data for retrieval purposes.

The choice of movies and television series as the project's central theme was motivated by their cultural significance, the abundance of available data, and the diversity of attributes they encompass. This domain offers a rich combination of structured data, such as titles, genres, release years, and ratings, alongside unstructured textual information, including plot summaries and user-generated reviews. These characteristics make it a valuable case study for exploring the interaction between metadata and natural language content within information retrieval systems.

Despite the magnitude of available data in this area, existing information retrieval systems often remain limited, focusing primarily on keyword-based searches or rigid metadata filters. Such systems fail to explore the deeper semantic and contextual connections within textual content, leaving room for improvement in the way users discover and relate media content. This project aims to address this limitation by constructing a coherent, high-quality dataset that integrates structured and unstructured sources, establishing the foundation for more expressive retrieval approaches in later stages.

This document is structured in the following way: **Data Extraction and Sources** presents the data sources and extraction process. **Data Preparation and Quality Assessment** details data preparation procedures and quality assessment, outlining the reproducible pipeline employed. **Exploratory Data Analysis and Characterization** explores the datasets through statistical and textual characterization. **Conceptual Model** introduces the conceptual model of the domain, followed by **Document Definition**, which defines the notion of documents adopted for information retrieval. **Information Needs and Search Scenarios** discusses potential information needs and search scenarios, and **Conclusions and Future Work** concludes with reflections and directions for future work.

2 Data Extraction and Sources

When selecting the dataset to be used in this project, we chose **IMDb (Internet Movie Database)** [1] as our primary source. IMDb offers a vast collection of movies and series, providing numerous attributes for each title, such as release year, genre, cast, and ratings. In addition, it serves as one of the most widely recognized platforms for user-generated reviews and ratings, making it a reliable and comprehensive data source.

The information provided is available through various structured non-commercial datasets that can be freely accessed for research purposes, with the exception of movie and series descriptions. To address this limitation, web scraping was employed using data from the **StreamWithVPN** [2] website, which provided the missing textual descriptions. This complementary source ensured a complete dataset by filling the descriptive gap present in IMDb's publicly available data.

For this work, the following datasets were used: the IMDb core files (`title.basics`, `title.ratings`, `title.principals`, and `name.basics`), supplemented with the extracted textual descriptions from StreamWithVPN.

Table 1: Summary of Movie/Series datasets.

Dataset	Features	Entries	Size (MBs)
Title Basics IMDb	9	11962455	987.0
Title Ratings IMDb	7	1619879	26.9
Title Principals IMDb	6	95180101	3950.0
Name Basics IMDb	6	14742901	864.0

Was also considered it relevant to include movie and series reviews to further enrich our data collection. For this purpose, we selected two different datasets: one from **Kaggle**[3] for movie reviews and another from **OpenDataBay**[4] for series reviews. Reviews directly from IMDb were not used because they are not provided in the available IMDb datasets. The chosen datasets are presented below:

Table 2: Summary of the Reviews datasets.

Dataset	Features	Titles	Reviews	Size (MBs)
Movie Reviews	8	15951	664436	308.28
Series Reviews	13	3136	52895	38.5

3 Data Preparation and Quality Assessment

The dataset construction followed a rigorous and reproducible pipeline (Figure 3) inspired by standard information retrieval methodologies. The consolidated dataset was obtained by integrating IMDb core files—`title.basics`, `title.ratings`, `title.principals`, and `name.basics`—with enriched textual data gathered through targeted web scraping. Intermediate results were exported in TSV format to ensure traceability and reproducibility throughout the process.

The **initial cleaning phase** consisted of the systematic removal of incomplete, null, or uninformative records to ensure dataset integrity. Subsequently, categorical restrictions were applied, limiting the corpus to `feature films` and `television series` while excluding other title categories. Entries lacking essential rating information were discarded, and a threshold of 1,000 minimum votes was imposed to mitigate sparsity and popularity bias. To further refine representativeness, a Bayesian weighted-rating formula was implemented, using the global mean rating and the 75th percentile of vote counts as priors, resulting in a ranked subset of 10,000 titles for enrichment.

The **normalization phase** addressed structural and textual inconsistencies across datasets. Titles were standardized by removing extraneous metadata such as release years or additional descriptors. For instance,`Dekalog (1988)` was normalized to `Dekalog`, and `The Office (UK): Season 3` to `The Office`, ensuring uniformity across entries. This process reduced variability and improved cross-source matching accuracy. Similarly, non-standard delimiters and encoding inconsistencies were resolved to achieve a unified schema.

Given that the reviews dataset originated from a distinct source, a **matching procedure** was required to associate reviews with the correct movie or series entities. This alignment employed the `fuzzywuzzy`[5] library, applying an 80% similarity threshold to reconcile discrepancies caused by non-printable characters and hidden formatting. Upon successful matching, the “title” column in the reviews dataset was replaced by the unique IMDb identifier (**tconst**), adopted as a foreign key to enable consistent cross-referencing between reviews and titles.

Further **enrichment procedures** involved joining principal-role records with corresponding person metadata to extract up to three unique actors or actresses per title along with their respective character names. All string attributes were cleaned to remove escape sequences and malformed arrays. Web scraping was conducted in batch mode with retry logic and polite delays to acquire textual descriptions of movies and series. Each operation was logged to maintain an auditable record of success and failure rates.

The **final normalization and validation** steps included deduplication based on unique identifiers, profiling of missing values, and standardization of data formats, such as enforcing canonical genre delimiters. The resulting dataset exhibited high structural coherence and consistency across all attributes.

Despite extensive preprocessing, **known issues** persisted. A minor subset of titles lacked descriptions due to failed retrieval attempts, and occasional parsing errors were observed in third-party metadata fields. Additionally, certain user-generated texts contained non-ASCII characters that were retained as NaN values for transparency. These anomalies were documented and preserved in compliance with best practices for data integrity and reproducibility.

4 Exploratory Data Analysis and Characterization

This section presents a comprehensive analysis of the processed dataset, examining multiple dimensions to understand the characteristics and patterns within movie and series reviews. The graphs and tables were obtained using the Matplotlib [6], Numpy [7] and NLTK [8] libraries.

4.1 Descriptive Analysis

From the IMDb dataset, we selected the 10,000 best-rated titles, comprising 3,216 series and 6,784 movies. Among these, we successfully retrieved plot descriptions for 7,976 titles through web scraping, achieving a 79.76% success rate. This process ensured that the majority of our selected entries contained comprehensive textual metadata suitable for further analysis.

Regarding the review datasets, the s Reviews dataset originally included 52,895 reviews across 3,136 unique titles. From these, we successfully matched 1,038 series titles to our selected IMDb subset, resulting in a total of 28,920 corresponding reviews, resulting in a correspondence rate of 54, 90%. Similarly, in the Movie Reviews dataset, which contained 664,436 reviews, we were able to link 5,532 movie titles to our selected IMDb movies, which represents 358,152 reviews with a correspondence rate of 53, 9%.

In general, the data scraping process to obtain plot descriptions was satisfactory, achieving nearly 80% success. The correspondence between the review datasets and the selected IMDb titles was also acceptable, with more than half of all reviews successfully matched to the chosen titles. These results indicate that the integrated dataset maintains a solid level of completeness, combining both structured information and review-based content suitable for subsequent analysis.

4.2 Review Distribution Analysis

Analysis of the most reviewed titles (Figure 4) reveals superhero and franchise films dominating user engagement. Zack Snyder’s Justice League leads with approximately 1,800 reviews, followed by Wonder Woman and The Dark Knight Rises with over 1,600 reviews each. This pattern suggests heightened engagement with blockbuster releases and established franchises, which typically generate more polarized opinions and discussion.

The correlation analysis between ratings and review counts (Figure 5) yields a coefficient of 0.03, indicating virtually no linear relationship between a title’s rating and its review volume. This finding suggests that review quantity is driven more by factors such as marketing reach, franchise popularity, and cultural impact rather than perceived quality.

4.2.1 Genre Distribution Analysis. The genre distribution analysis (Figure 6) demonstrates clear patterns in review engagement across different content categories. Science Fiction leads with an average of 173 reviews per title, followed by Adventure (139) and Action (133). In contrast, niche genres such as Talk-Show, Reality-TV, and Short Films average fewer than 35 reviews per title.

Interestingly, when examining average ratings by genre, an inverse pattern emerges. Short Films achieve the highest average rating (8.27), followed by Talk-Shows (8.26) and Game-Shows (8.15), while mainstream genres typically receive lower average scores. This suggests specialized content attracts more selective, appreciative audiences, while mass-market genres face broader, more critical viewership.

These analytical results provide valuable guidance for developing our system as understanding how engagement and ratings vary across genres enables more informed data structuring and feature prioritization.

4.3 Analysis of Word Clouds

To acquire a better understanding of the key terms and themes within our datasets we generated the following word clouds:

4.3.1 Description Word Cloud (Figure 7). The first word cloud, generated from the collection of movie /series descriptions in our dataset reflects the thematic and narrative structure of the movies and series themselves. Words like life, world, family, love, find, and live dominate, pointing to recurring motifs around human relationships, personal struggles, and existential experiences. The prominence of terms such as man, woman, son, father, and friend underscores the centrality of interpersonal dynamics in most story lines.

4.3.2 Reviews Word Cloud (Figure 8). The second word cloud highlights the most frequent terms used by users when expressing opinions about movies and series. Words such as movie, film, good, great, story, and characters dominate, suggesting that user discussions are primarily centered on the overall quality of storytelling, character development, and general enjoyment. Positive sentiment terms like best, love, and amazing also appear frequently, indicating a generally engaged and expressive audience that emphasizes emotional and qualitative evaluations rather than technical aspects.

5 Conceptual Model

The conceptual model represents the logical organization of the data used in this project, serving as the foundation for subsequent information retrieval and analysis tasks. It establishes how different entities interact within the domain of movies and television series.

As illustrated in Figure 2, the dataset is composed of two main entities: **Movie/Series** and **Review**. The *Movie/Series* entity stores structured metadata collected from authoritative sources, including identifiers, titles, release years, genres, ratings, cast information, and textual descriptions. The *Review* entity encapsulates unstructured textual data written by users, associated with the corresponding movie through the attribute `review_tconst`.

This relationship is an optional one-to-many association, where each movie or series can be associated with zero or more reviews, and each review is related to exactly one movie or series. This structure allows flexible querying, supporting retrieval at both the document level (e.g., specific reviews) and the item level (e.g., aggregated movie information).

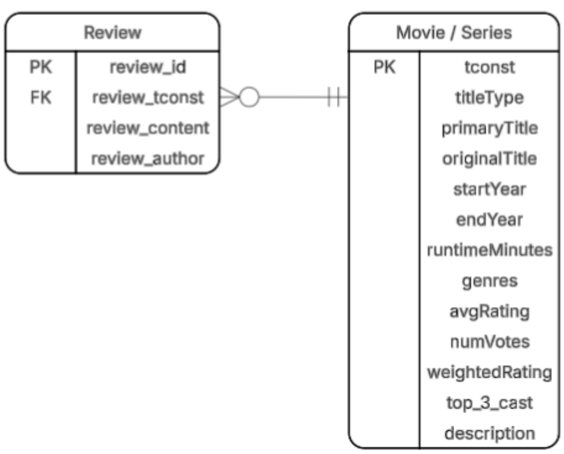

Figure 2: Conceptual Model

6 Document Definition

In our search result document, we include the following elements:

- Movie/Series ID: Unique identification string that represents the movie/series
- Title Type: Distinguishes if the entry is a movie or a series
- Primary Title: Title of the movie/series in English
- Original Title: Title of the movie/series in the original language (if it applies)
- Start Year: movie/series release year
- End Year: Series release year (movies don't apply)
- Runtime: Time the movie lasts (or average time of a series' episode)
- Genres: List of genres that can describe the movie/series
- Average Rating: Rating given by the people to the movie/series in IMDB
- Number of Votes: The amount of votes/ratings a movie/series has
- Top 3 Cast: The top 3 cast members in the movie/series and the respective characters they play
- Description: Text that describes the movie/series (most around 200/300 words)

We also have separate result documents for the reviews, since each movie/series has multiple reviews. These are the elements in them:

- Review ID: Unique identification string that represents the review
- Reviews tconst: Unique Identification from related Movie/Series
- Review Content: Text expressing author's opinion
- Review Author: Author of the review

7 Information Needs and Search Scenarios

To design an effective search engine for movies and series, it's important to understand users' information needs. Below are several key types of queries that reflect different aspects of the system's functionality.

- Similarity Search: "Find movies similar to Inception"
- Theme Based Discovery: "Retrieve movies about artificial intelligence"
- Era and Genre Filtering: "List comedies from the 90s"
- Runtime Finder: "Get movies with a runtimes below 90 minutes"
- Actor Based Search: "Series where Bryan Cranston is featured"
- Rating-Weighted Discovery: "Best reviewed content during pandemic era (2020-2022)"
- Reviewer Trust Metrics: "Films recommended by Roger Ebert"
- Length-Aware Review Filtering: "Movies with detailed analytical reviews (average review above 500 words)"
- Cross-Media Recommendations: "Films similar to Game of Thrones but standalone"

8 Information Retrieval

Information Retrieval represents the systematic process of discovering and extracting pertinent information from extensive collections of inherently unstructured data sources, particularly textual content[9]. This extraction operates through the use of structured documents—derived from the reorganization of raw data—with results ranked according to their relevance to user queries, which constitutes the primary challenge in this domain.

This section outlines the indexing and querying methodologies employed in the cinematic data retrieval system, leveraging the previously constructed movie and series documents. The search system implementation utilizes Apache Solr[10], an open-source platform offering numerous capabilities aligned with the project's objectives, including distributed and efficient indexing, horizontal scalability, and sophisticated search functionalities that extend beyond simple full-text matching to enable semantic and contextual retrieval of movie and television content.

8.1 Document Characterization

The documents subject to indexing and retrieval operations represent the output of the data extraction, enhancement, and consolidation stages outlined in the previously described pipeline.

Consequently, each movie document encompasses multiple attributes: primary and original titles, content type classification, temporal information (release and end years), duration, genre classifications, rating metrics (mean rating, vote count, weighted rating), principal cast members, and plot synopsis.

8.2 Indexing Process

The indexing mechanism constitutes a critical component of Information Retrieval systems, facilitating efficient query processing through systematic data organization. This process establishes a well-structured index that dramatically improves retrieval speed and system scalability. In the absence of appropriate indexing strategies, retrieval systems would encounter significant performance degradation, manifesting as prolonged response latencies and elevated computational demands.

Solr implements diverse indexing methodologies for document attributes and their corresponding queries, leveraging Tokenizers [11] and Filters [12]. Tokenizers decompose input strings into token sequences according to specified rules, while Filters refine these tokens to ensure uniformity across search operations and pattern matching.

For this implementation, distinct indexing strategies were employed based on field characteristics and search requirements. Title attributes necessitate prefix-based matching to enable autocomplete functionality, cast member names require phrase-preserving analysis, and plot descriptions demand comprehensive text search with semantic expansion.

Several specialized field types were configured to address varying search scenarios:

'text_general' designed for descriptive content:

- ***StandardTokenizerFactory*** tokenizer: segments text using whitespace and punctuation delimiters;
- ***LowerCaseFilterFactory*** filter: transforms all characters to lowercase representation;
- ***ASCIIFoldingFilterFactory*** filter: normalizes diacritics and special characters to ASCII equivalents;
- ***StopFilterFactory*** filter: eliminates frequently occurring English function words;
- ***EnglishPossessiveFilterFactory*** filter: strips possessive suffixes;
- ***PorterStemFilterFactory*** filter: applies stemming to derive word roots, enabling retrieval of morphological variants;

'title_text' configured for title attributes supporting prefix matching:

- ***StandardTokenizerFactory*** tokenizer: applies standard text segmentation;
- ***LowerCaseFilterFactory*** and ***ASCIIFoldingFilterFactory*** filters: perform character normalization;
- ***WordDelimiterGraphFilterFactory*** filter: processes compound terms and camelCase notation;
- ***EdgeNGramFilterFactory*** filter: generates character sequences (4-8 length) from token prefixes during index construction;

'name_text' tailored for actor name processing:

- ***StandardTokenizerFactory*** tokenizer combined with normalization filters;
- ***ShingleFilterFactory*** filter: constructs multi-token units (2-3 word sequences) to maintain name phrase integrity during indexing;

'string_ci' implemented for case-insensitive categorical attributes:

- ***KeywordTokenizerFactory*** tokenizer: preserves complete field value as atomic token;
- ***LowerCaseFilterFactory*** and ***TrimFilterFactory*** filters: apply minimal normalization;

Query analysis incorporates **SynonymGraphFilterFactory** for plot descriptions and title fields, which proves essential given the diverse terminology employed in cinema-related searches (e.g., "movie" versus "picture", "cast" versus "actors").

Copy field declarations were established to enable faceting and sorting capabilities by replicating indexed textual fields to non-tokenized string variants with docValues activation.

The complete document indexing configuration follows this schema:

Table 3: Schema Field Types

Field	Type	Indexed
primaryTitle	title_text	yes
originalTitle	title_text	yes
titleType	string_ci	yes
startYear	pint	yes
endYear	pint	yes
runtimeMinutes	pint	yes
genres	string_ci	yes
averageRating	pfloat	yes
numVotes	pint	yes
weightedRating	pfloat	yes
top_3_cast	name_text	yes
top_3_cast_search	text_general	yes
description	text_general	yes

8.3 Retrieval Process and Setup

The implementation strategy encompasses two distinct schema configurations: a baseline schema employing Solr's native field types for all attributes, and an enhanced schema leveraging custom-defined field types to improve search performance.

Both schema configurations share common query parameters:

- **Query (`q`)**: Targets the most semantically significant terms within the user's query.
- **Query Operator (`q.op`)**: Specifies boolean logic using OR | AND operators for term combination.
- **Query Filter (`fq`)**: Establishes constraints to narrow the candidate document set.

- **Filter List (`fl`)**: Controls which fields appear in the query response.
- **Sort Field (`sort`)**: Determines the ordering criterion for result presentation.

Table 4: Query parameters

Parameter	Value
q	(action thriller)
q.op	OR
fq	genres:"Action" AND startYear:[2010 TO *]
fl	primaryTitle, averageRating, genres, startYear
sort	weightedRating desc

The baseline schema relies on Solr's default field type handling, whereas the optimized schema utilizes the eDismax query parser [13] with specialized parameters designed to refine retrieval effectiveness:

- **Query Field with Optional Boost (`qf`)**: Distributes differential weights across searchable fields to prioritize certain content;
- **Phrase-Boosted Field (`pf`)**: Emphasizes documents containing query terms in proximity;
- **Phrase Boost Slope (`ps`)**: Specifies the maximum positional distance permitted between query terms for phrase matching;

Table 5: defType eDismax parameters

Parameter	Value
qf	primaryTitle^2 genres^2 description^3 top_3_cast^2
pf	description^6
ps	3

The differential weighting scheme within the **qf** parameter establishes a hierarchical importance: **primaryTitle** receives highest priority as users frequently search by movie name, followed by **originalTitle** for international films, **description** for plot-based queries, and **top_3_cast_search** for actor-driven searches. The **pf** parameter concentrates additional weight on **primaryTitle** and **description** fields, promoting documents where query terms appear as cohesive phrases. This configuration pairs with a **ps** value of 2, accommodating typical spacing patterns in movie titles and descriptions.

The eDismax parameter **bq** (boost query) underwent experimental evaluation but was ultimately excluded from the production system. Its query-specific nature would introduce inconsistency across different information needs, compromising the system's generalizability and potentially skewing results based on individual query characteristics.

Maintaining this enhanced retrieval strategy across all query types has strengthened the system's search capabilities, yielding measurable improvements in result quality as demonstrated in the evaluation section.

9 Evaluation

Evaluation constitutes a critical component of Information Retrieval systems, dependent on the target document corpus and the nature of information requirements. Understanding user search patterns is essential for iterative system refinement based on performance feedback. For this evaluation, emphasis was placed on effectiveness—the system's capability to retrieve relevant information—rather than efficiency, which concerns retrieval speed.

Relying solely on subjective or individual assessment metrics could introduce bias when comparing the baseline and optimized schema implementations. To ensure objectivity, a comprehensive set of metrics grounded in **precision** and **recall** were employed, including **Average Precision (AvP)**, **Precision at K (P@K)**, **Precision-Recall curves**, and **Mean Average Precision (MAP)**. Precision quantifies the proportion of truly relevant documents among retrieved results, while recall measures this ratio relative to all relevant documents in the collection. Given the collection contains 10,000 movie documents, exhaustive relevance assessment proved impractical, necessitating manual evaluation based on the top twenty retrieved documents per query.

The **Average Precision (AvP)** metric holds particular significance as precision directly correlates with user satisfaction in most search scenarios. Users typically prioritize the relevance of top-ranked results over exhaustive recall, since the total number of relevant documents in the system remains unknown during normal usage, unlike the immediate visibility of initial search results. For **Precision at K (P@K)**[14], evaluation focused on the top twenty documents per query, representing a realistic threshold that aligns with typical user browsing behavior in search interfaces.

Precision-Recall Curves were constructed for each query across both schema configurations using the ranked subset of returned documents. System stability is indicated by curve smoothness, while overall performance is quantified through the Precision-Recall Area Under the Curve [15] [16]. This metric captures the system's holistic effectiveness in balancing precision and recall across varying retrieval thresholds.

The **Mean Average Precision (MAP)** serves as a standard Information Retrieval metric representing the mean of Average Precision values across multiple query evaluations. This metric reveals system consistency when addressing diverse information needs, providing insight into generalization capability beyond individual queries [17].

The evaluation leveraged TREC evaluation tools and methodologies, with relevance judgments (qrels) manually created for three distinct query scenarios. Query results from both schemas were converted to TREC format using custom scripts (`solr2trec.py`), enabling standardized metric computation through `trec_eval`. Precision-Recall curves were generated using visualization scripts (`plot_pr.py`) to facilitate comparative analysis.

In the subsequent subsections, three representative user scenarios are presented as queries, each accompanied by retrieval results and statistical analysis based on precision and recall metrics for both schema configurations.

A *'Funny TV Show'*

Information Need:

- **Query 1 - Baseline:** Basic search for funny TV shows from 2000-2015. Searches across title, description, genres, and cast with equal weights.
- **Query 1 - Boosted:** Same search but prioritizes description (3x boost) and boosts title, genres, and cast (2x each). Adds phrase boosting (6x) for exact phrase matches in descriptions with 3-word slop tolerance.

Table 6: Q1 Basic Schema

Rank	Baseline	Boosted
P@20	0.7	0.75
AP	0.87	0.88

Table 7: Q1 Intermediate Schema

Rank	Baseline	Boosted
P@20	0.8	0.8
AP	0.92	0.94

Result Analysis:

- **Basic Schema Performance:**
  As depicted in Figure 9 and Table 6, baseline maintains 100% precision until 30% recall, then drops to 90% at 40% recall and 75% at full recall. Boosted version shows gentler decline, maintaining 90% through 60% recall before dropping to 80%. The Boosted version shows marginal improvement with slightly better precision throughout.
- **Intermediate Schema Performance:**
  As seen in Figure 10 and Table 7 , both versions nearly identical, maintaining 100% precision until 45-50% recall, declining to 90% at 60% recall and 80-85% at full recall. Smoother curves with higher precision at high recall compared to basic schema. Both queries achieve identical P@20 but boosted version has better overall ranking quality.

Key Findings:

- **Schema Impact:** The intermediate schema achieves 80% P@20 compared to basic schema's 70-75%, showing that proper field configuration improves accuracy by 10%.
- **Boosting Effectiveness:** Field boosting improves ranking quality (higher AP) but has minimal impact on P@20. The main benefit is better ordering of relevant results rather than finding more relevant documents.

B *'Robert De Niro* or *Al Pacino* Crime Drama'

Information Need:

- **Query 2 - Baseline:** Basic search for highly-rated crime drama movies (rating 7.5+) starring Robert De Niro or Al Pacino. Searches across all fields with equal weights.
- **Query 2 - Boosted:** Same search but prioritizes description (3x boost) and boosts title, genres, and cast (2x each). Adds phrase boosting (6x) for exact phrase matches in descriptions with 3-word slop tolerance.

Table 8: Q2 Basic Schema

Rank	Baseline	Boosted
P@20	0.75	0.75
AP	0.93	0.93

Table 9: Q2 Intermediate Schema

Rank	Baseline	Boosted
P@20	0.75	0.75
AP	0.93	0.93

Result Analysis:

- **Basic Schema Performance:**
  Figure 11 and Table 8 shows that, both Baseline and Boosted versions show identical curves. Precision remains at 100% until 70% recall, then drops sharply to 85% recall, and gradually declines to 75% at full recall. The sharp drop indicates a clear boundary between highly relevant and marginally relevant documents. Boosting shows no improvement.
- **Intermediate Schema Performance:**
  Figure 12 and Table 9 reveals that, curves are identical to basic schema, maintaining 100% precision until 70% recall, dropping to 85% recall, and ending at 75% at full recall. No difference between schemas or boosting strategies, confirming that actor name searches don't benefit from advanced text analysis. Again, boosting does not provide measurable benefit.

Key Findings:

- **Schema Impact:** Both schemas achieve identical performance (75% P@20, 0.925 AP). Advanced text analysis provides no benefit for searching for the specific actor name and genre.
- **Boosting Ineffectiveness:** Field boosting has zero impact. Actor names are already highly specific, making field prioritization unnecessary.

C *'artificial intelligence,* AI, robots, androids, *machine learning, sentient machines'*

Information Need:

- **Query 3 - Baseline:** Basic search for AI/robot-themed movies using terms like "artificial intelligence", AI, robots, androids, and machine learning. Searches across all fields with equal weights.
- **Query 3 - Boosted:** Same search but prioritizes description (3x boost) and boosts title, genres, and cast (2x each). Adds phrase boosting (6x) for exact phrase matches in descriptions with 3-word slop tolerance.

Table 10: Q3 Basic Schema

Rank	Baseline	Boosted
P@20	0.9	0.9
AP	0.84	0.84

Table 11: Q3 Intermediate Schema

Rank	Baseline	Boosted
P@20	0.6	0.6
AP	0.82	0.82

Result Analysis:

- **Basic Schema Performance:**
  As demonstrated in Figure 13 and Table 10, both Baseline and Boosted versions show nearly flat curves at 90-95% precision across all recall levels. The unusually stable performance suggests that the query retrieves a well-defined set of AI/robot movies with minimal false positives, although the lower absolute precision (60% P@20) indicates a wider retrieval of marginally relevant results. Boosting shows no improvement.
- **Intermediate Schema Performance:**
  As the results in Figure 14 and Table 11 indicate, the curves remain flat at 90-95% precision until 70% recall, where it shows a minor dip to 90% recall. It stays at 90% until the end. The dramatically improved performance (90% P@20 vs 60% basic) demonstrates that advanced text analysis successfully filters out non-AI themed movies that contain generic "machine" references. Again, boosting does not provide measurable benefit

Key Findings:

- **Schema Impact:** Intermediate schema achieves 90% P@20 compared to basic's 60%, showing a dramatic 30% improvement. This is the largest gain across all queries, proving that advanced text analysis is critical for thematic/conceptual searches.
- **Boosting Ineffectiveness:** Field boosting has zero impact in both schemas. The AI/robot terminology is distinctive enough that field prioritization does not add value.

10 Conclusions and Future Work

Previously we concluded the data preparation phase, in which heterogeneous data from multiple sources were collected, cleaned, integrated, and analyzed to produce a coherent and reproducible dataset. The resulting collection combines structured metadata from IMDb with unstructured textual content from external reviews, forming a solid foundation for subsequent information retrieval experiments. In this new milestone, we also refined our decisions regarding dataset scope and representation. Notably, we chose not to include each movie's or series' corresponding reviews as standalone documents, focusing instead on leveraging descriptions as the primary source of unstructured content. This approach simplified the schema while maintaining consistency in the dataset's structure. Additionally, the synonyms list used for query expansion could be strengthened with a larger set of terms and more semantically related alternatives. Increasing the complexity of the information needs would also create more realistic retrieval scenarios and highlight the benefits of a custom schema and boosted queries. The next steps will involve enhancing the dataset and retrieval capabilities by incorporating individual reviews as additional documents, expanding and refining the synonyms list, and designing richer information needs. Further improvements will include introducing NLP techniques and applying sentiment analysis to both descriptions and reviews. These advancements will support the development of a more comprehensive and semantically aware information retrieval system.

Annexes

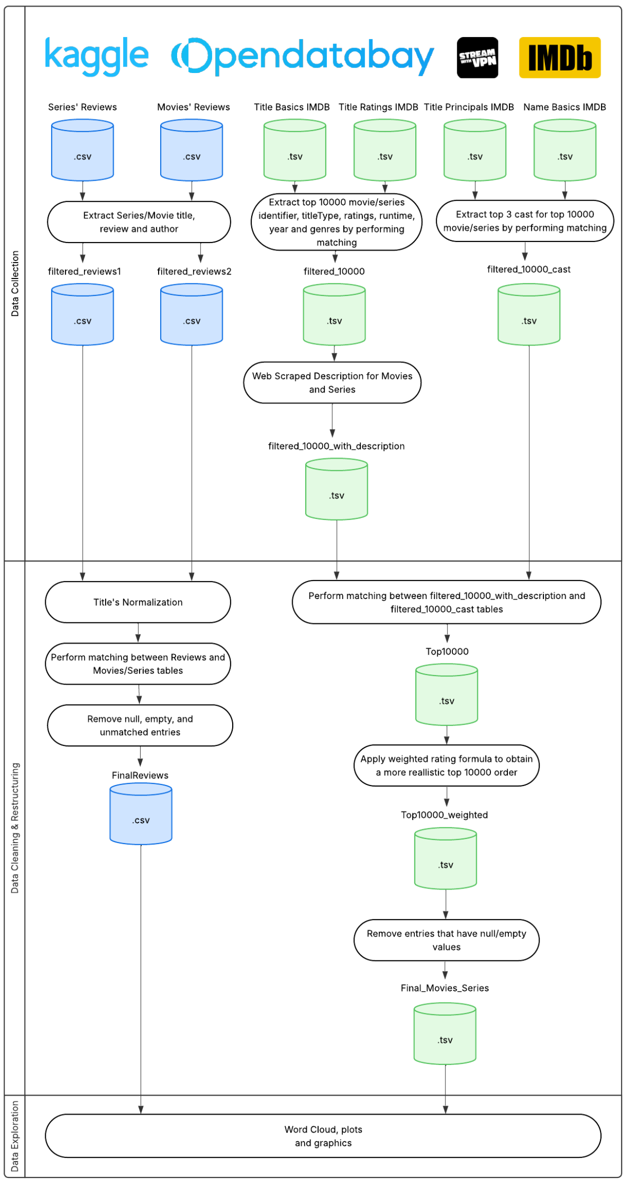

Figure 3: Pipeline

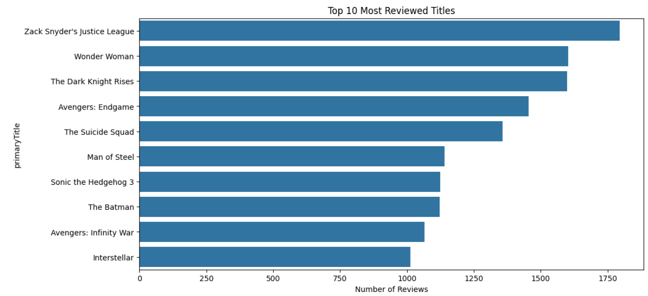

Figure 4: Most Reviewed Titles

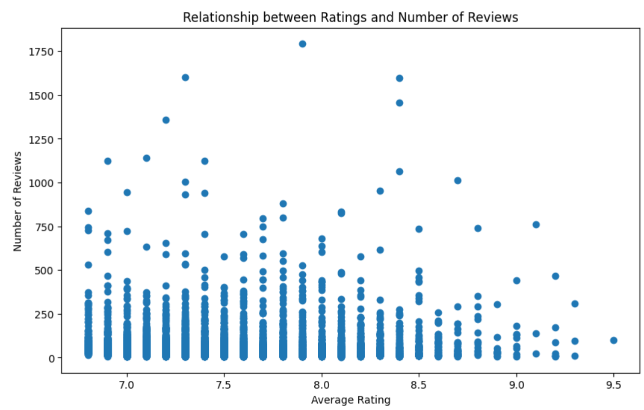

Figure 5: Distribution of Reviews per Title

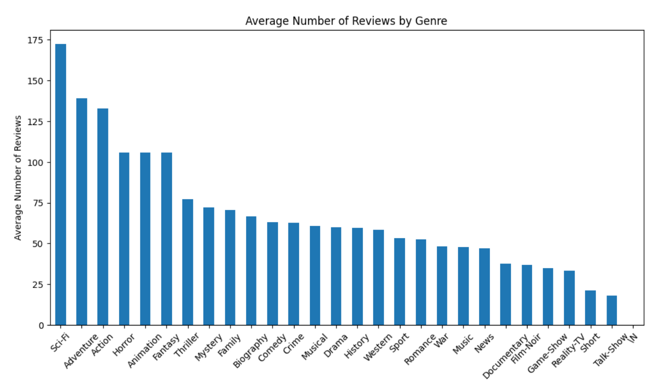

Figure 6: Genre Distribution of Reviews and Ratings

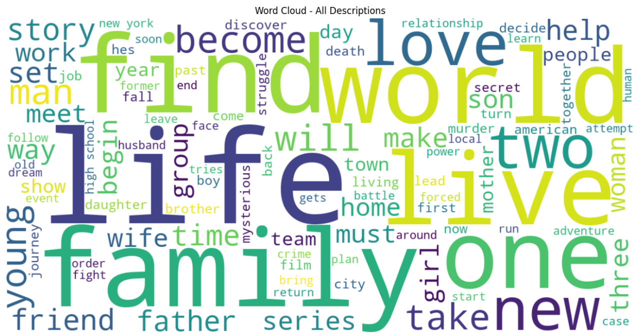

Figure 7: Description Word Cloud

Figure 8: Reviews Word Cloud

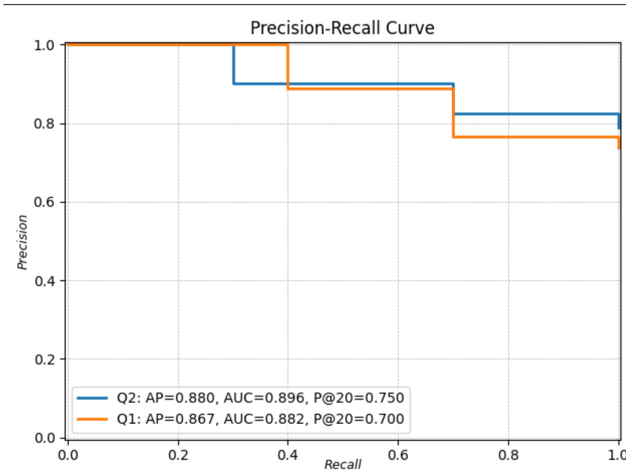

Figure 9: Query 1 - Basic

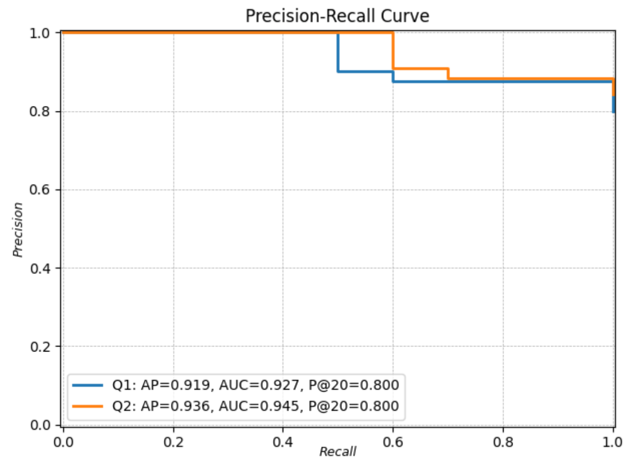

Figure 10: Query 1 - Intermediate

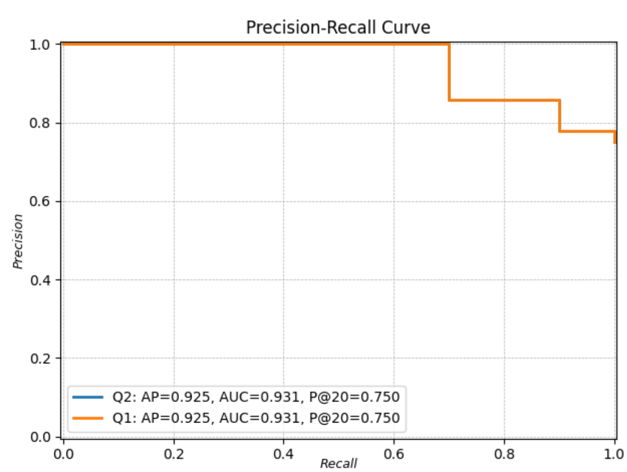

Figure 11: Query 2 - Basic

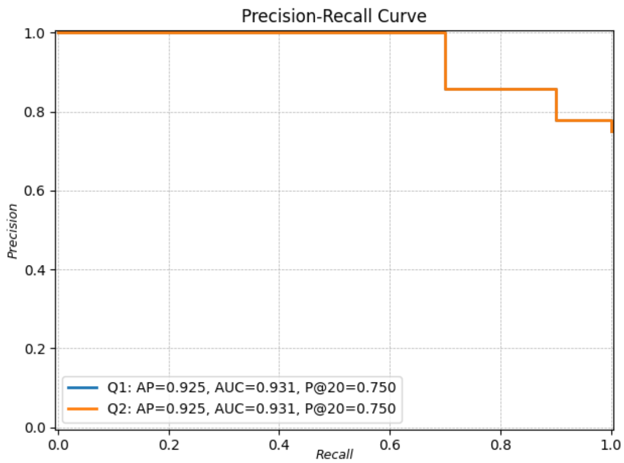

Figure 12: Query 2 - Intermediate

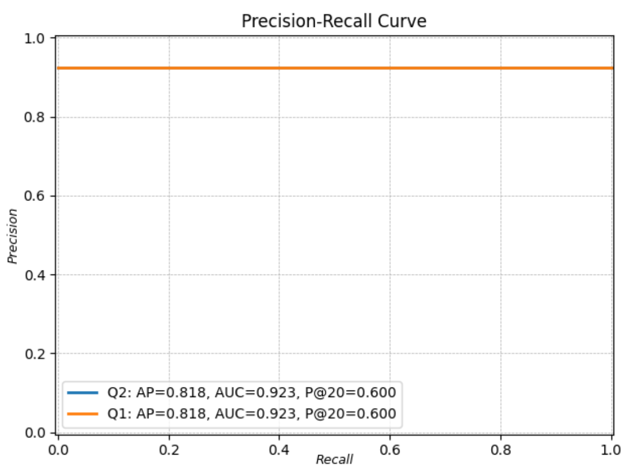

Figure 13: Query 3 - Basic

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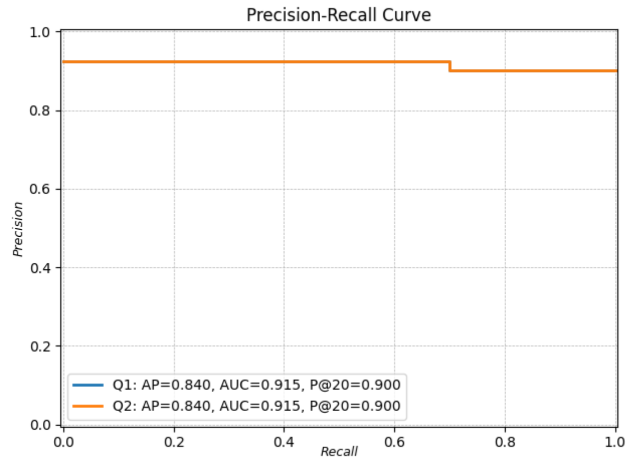

Figure 14: Query 3 - Intermediate

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